MATH 142: Elementary Algebraic Topology

By William Nguyen

Fall 2026

Contents

11/4/26 Lecture 29: Van Kampen's Theorem and Abstract Simplicial Complexes 1
Theorem 29.4: Van Kampen's Theorem 1
Definition 29.5: Finite Abstract Simplicial Complexes 3

11/4/26 Lecture 29: Van Kampen's Theorem and Abstract Simplicial Complexes

Here are some corollaries from the results proved last time.

Corollary 29.1

If K is a (finite) connected simplicial complex, then $\pi_1(|K|, v)$ is finitely presented.

Corollary 29.2

If $K(2)$ denotes the 2-skeleton, that is, the union of the vertices, edges, and triangles of K, then

$$\pi_1(|K|, v) \cong \pi_1(|K(2)|, v)$$

Corollary 29.3

If J and K are connected simplicial complexes that intersect at a point v, then

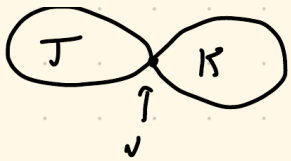
$$\pi_1(|J \cup K|, v) \cong \pi_1(|J|, v) * \pi_1(|K|, v)$$

Proof: Take maximal trees T_1 and T_2 in J and K respectively. Let $T_0 = T_1 \cup T_2$. This is a tree in $J \cup K$. Then, the generators are the edges in $J \setminus T_1$ ($\{g_{ij}\}$) and $K \setminus T_2$ ($\{h_{ij}\}$), and the relations come from 2-simplices in J ($\{r_p\}$) or K ($\{s_q\}$). Then,

$$\pi_1(|J \cup K|, v) = \langle \{g_{ij}\} \cup \{h_{ij}\} \mid \{r_p\} \cup \{s_q\} \rangle = \pi_1(|J|) * \pi_1(|K|)$$

■

Example: Two Circles



$$\pi_1(J \cup K) = \mathbb{Z} * \mathbb{Z} = \pi_1(|J|) * \pi_1(|K|)$$

Theorem 29.4: Van Kampen's Theorem

Suppose J, K are simplicial complexes, $J \cap K$ is a simplicial complex, and all of these are path connected. Choose $v \in J \cap K$. Let $j : J \cap K \to J$ and $k : J \cap K \to K$ be the inclusion maps. Then, $\pi_1(|J \cup K|, v)$ can be obtained from $\pi_1(|J|, v) * \pi_1(|K|, v)$ by adding the relations

$$j_*(z)k_*(z)^{-1}$$

where z runs over the generators of $\pi_1(|J \cap K|, v)$.

Proof: Choose a maximal tree T_0 of $J \cap K$. Extend to maximal trees T_1, T_2 in J, K respectively. Note $T_1 \cup T_2$ is a maximal tree in $J \cup K$.

Then, $\pi_1(|J \cup K|, v)$ is generated by the edges in $J \cup K$ not in $T_1 \cup T_2$, which are

$$\{\text{edges in } J \setminus T_1\} \cup \{\text{edges in } K \setminus T_2\}$$

but there are some redundancies for edges in $(J \setminus T_1) \cap (K \setminus T_2)$, so we add the relations

$$a_{ij} = b_{ij}, \quad a_{ij} \in G(J, T_1),\ b_{ij} \in G(K, T_2)$$

for edges $v_i v_j$ in $(J \cap K) \setminus T_0$.

■

Special Case: If J, K are simply connected and $J \cap K$ is nonempty and path connected, then $\pi_1(J \cup K, v)$ is a quotient of

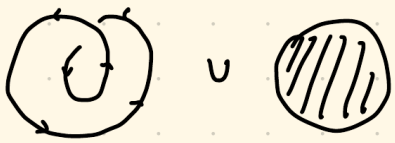
$$\pi_1(J, v) * \pi_1(K, v) = \{e\}$$

so $J \cup K$ is simply connected.

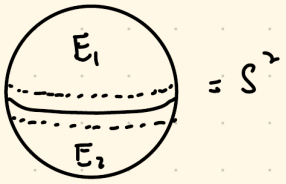
Example: The Projective Plane

Consider $P^2 = S^2/\mathbb{Z}_2$. We know $\pi_1(P^2) = \mathbb{Z}_2$ since $\pi_1(S^2)$ is trivial and $S^2 \to P^2$ is a covering map. We will show this with Van Kampen.

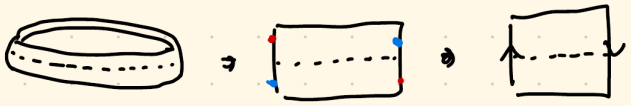
Claim: $P^2 = \{\text{Möbius strip}\} \cup_{S^1} D^2$



The reason is that if



then elements in E_1 are identified with elements of E_2, so these can be represented by $E_1 \cong D^2$. For elements of the equator, these are locally identified by the Möbius strip:



Then, let $J =$ Möbius strip, $K = D^2$, $J \cap K = S^1$, and $J \cup K = P^2$. Thus, by Van Kampen, and from the fact that the inclusion of a generator of $\pi_1(S_1)$ into the Möbious strip is a loop that runs around twice, we see

$$\begin{aligned}\pi_1(P^2) &= \frac{\pi_1(\text{Möbius strip}) * \pi_1(D^2)}{\langle j_*(z)k_*(z)^{-1}\rangle} \\ &= \frac{\langle z\rangle * \langle e\rangle}{\langle z^2\rangle} \\ &= \mathbb{Z}/2\mathbb{Z}\end{aligned}$$

Definition 29.5: Finite Abstract Simplicial Complexes

A *finite abstract simplicial complex* is a pair (V, S) where V is a finite set and S is a collection of nonempty subsets of V such that

(i) $\{v\} \in S$ for all $v \in V$

(ii) $\mathcal{P}(X) \setminus \{\emptyset\} \subseteq S$ for all $X \in S$

Theorem 29.6

Any finite abstract simplicial complex can be realized as a simplicial complex in $\mathbb{R}^n$ for some n.

Proof: Say V has k elements. Let Δ be the $(k-1)$-dimensional simplicial complex corresponding to

$$\left\{x \in \mathbb{R}^k \,:\, x_i \geq 0,\ \sum_{i=1}^{k} x_i = 1\right\}$$

Any collection of vertices in Δ spans a simplex. Thus, if $V = \{v_1, \ldots, v_k\}$, match these to the vertices $\{e_1, \ldots, e_k\}$ of Δ, so that for each $X \in S$, we see that the vertices corresponding to the elements of X form a simplex. Do this for all $X \in S$ and we get a simplicial complex.

■

Example: Realizing an Abstract Complex

$V = \{a, b, c\}$, $S = \{\{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}\}$. Then, putting this in $\mathbb{R}^3$, we see the corresponding simplicial complex is

$$V' = \{e_1, e_2, e_3\}, \quad S' = \{\{e_1\}, \{e_2\}, \{e_3\}, \{e_1, e_2\}, \{e_1, e_3\}\}$$

